

The Transformer Revolution

GPT. Claude. Gemini. LLaMA. T5. BERT. SBERT. Stable Diffusion (with Cross-Attention). Every one is a transformer descendant. Today we open the hood.

Which of the LLMs above is encoder-only? Hold your guess — reveal at slide 10.

WHAT'S INSIDE

- 01 Why recurrence had to go
- 02 Self-attention — weighted average of...
- 03 Why scaled — why $\sqrt{d}$
- 04 Positional encoding — the order signal
- 05 Three flavours

WHY THIS LESSON MATTERS

One 2017 paper. Every modern LLM. 1.7 trillion parameters.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Why RNNs were replaced by transformers in 2017.

2

Self-attention as a learned weighted average.

3

Reason about Q, K, V and scaled dot-product.

4

Explain why positional encoding is needed.

5

Distinguish encoder-only, decoder-only, enc-dec.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Why recurrence had to go

Parallelism + long-range dependencies

02

Self-attention — weighted...

Each token asks every other token how relevant it is

03

Why scaled — why $\sqrt{d}$

Numerical stability

04

Positional encoding — the...

Without it, "Dog bites man" == "Man bites dog"

01

PART 1 OF 4

Why recurrence had to go

Parallelism + long-range dependencies

Why recurrence had to go

RNNs process tokens sequentially — cannot parallelise.

Long-range dependencies decay with distance.

Gradient flow fragile at depth.

Transformers: all tokens processed at once; direct long-range links.

Attention Is All You Need

Vaswani et al., NeurIPS — the cited line.

DEFINITION

Why recurrence had to go

Parallelism + long-range dependencies

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Self-attention — weighted average of tokens

Every token projects into Q, K, V.

$\text{attention} = \text{softmax}(QK^T / \sqrt{d}) \cdot V$.

QK^T measures compatibility between all pairs.

Softmax turns compatibilities into weights summing to 1.

ONE LINE

$\text{softmax}(QK^T / \sqrt{d})V$

The entire transformer trick, in one formula.

KEY POINTS

- Every token projects into Q, K, V
- $\text{attention} = \text{softmax}(QK^T / \sqrt{d}) \cdot V$
- QK^T measures compatibility between all pairs
- Softmax turns compatibilities into weights summing to 1

CONCEPT 3 OF 4

Why scaled — why $\sqrt{d}$

Large $d \rightarrow$ huge dot-products $\rightarrow$ saturated softmax.

Saturated softmax $\rightarrow$ vanishing gradients.

Divide by $\sqrt{d} \rightarrow$ back to reasonable scale.

Small detail; huge training-stability impact.

STABILITY

/ $\sqrt{d}$

Do not skip. The original paper credits this for training success.

DEFINITION

Why scaled — why $\sqrt{d}$

Numerical stability

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Positional encoding — the order signal

Transformers have no recurrence, no order by default.

Add a sinusoidal signal to each token's embedding.

$$\text{PE}(\text{pos}, 2i) = \sin(\text{pos} / 10000^{(2i/d)})$$

$$\text{PE}(\text{pos}, 2i+1) = \cos(\text{pos} / 10000^{(2i/d)})$$

ELEGANT

sin + cos

Different frequencies carry different positional scales.

KEY POINTS

- Transformers have no recurrence, no order by default
- Add a sinusoidal signal to each token's embedding
- $\text{PE}(\text{pos}, 2i) = \sin(\text{pos} / 10000^{(2i/d)})$
- $\text{PE}(\text{pos}, 2i+1) = \cos(\text{pos} / 10000^{(2i/d)})$

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Compute scaled dot-product attention by hand

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 Compute scaled dot-product attention by hand
- 3 Best pair shows on the board.
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Why did transformers take 17 years to arrive after RNNs?

Q2

Which variant will your next project use — and why?

Q3

Where would a transformer be overkill?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Year transformers were introduced:

A

B

B

D

C

Answer: C. Vaswani et al., NeurIPS 2017.

D

(no option)

Year transformers were introduced:



CORRECT ANSWER

D

WHY

The correct option matches the lesson's core principle.

Why divide by $\sqrt{d}$?

A

Speed

B

Stability of softmax

C

Privacy

D

Memory

Why divide by $\sqrt{d}$?

CORRECT ANSWER



Stability of softmax

WHY

Large dot-products saturate softmax; dividing rescales

BERT is:

A

Encoder-only

B

Decoder-only

C

Encoder-decoder

D

RNN

BERT is:



CORRECT ANSWER

Encoder-only

WHY

BERT is an encoder-only transformer used for classification / embeddings

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

RNNs process tokens sequentially — cannot parallelise.

Long-range dependencies decay with distance.

Gradient flow fragile at depth.

Transformers: all tokens processed at once; direct long-range links.

Attention Is All You Need

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THE TAKEAWAY

"softmax($QK^T/\sqrt{d}$)V. Plus positional encoding. That is the entire trick."

WHAT TO NOTICE

1

Why recurrence had to go

2

Self-attention — weighted...

3

Why scaled — why $\sqrt{d}$

4

Positional encoding — the...

Mini assignment — explain to a 10-year-old

Read the first 5 pages of "Attention Is All You Need".

DELIVERABLES

- Write 3 sentences explaining attention to a 10-year-old.
- Post on the cohort forum.
- Vote on the two best explanations.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

Deliverable: 3-sentence explanation on the forum.

ONE THING TO REMEMBER

“

"softmax($QK^T/\sqrt{d}$)V. Plus positional encoding. That is the entire trick."

— AI Learning Hub

END OF LESSON 4

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — Time-Series Forecasting. Your first Prophet model.